

JINGYAO YU

✉ peteryu0131@gmail.com | ☎ (778) 814-8601 | 🔗 [linkedin.com/in/jingyao-yu](https://www.linkedin.com/in/jingyao-yu) | 🐙 github.com/peteryu0131
🌐 https://peteryu0131.github.io/Jingyao_Yu_Website/

EDUCATION

MSc in Computer Science

September 2025 – Present

University of Victoria

GPA: 8.67/9.00

B.S. in Computer Science

January 2021 – July 2025

University of Victoria

RESEARCH INTERESTS

Simulation & Scientific ML: Differentiable simulation, physics-based modeling, scientific machine learning.

Vision & Geometry: Computer vision, 3D understanding, computer graphics.

AI & Interaction: AI agents, human–AI interaction, AI-assisted systems.

SUBMITTED MANUSCRIPT

Nafiseh Izadyar, **Jingyao Yu**, and Teseo Schneider. *GLANCE: Free-Form 3D Object Identification through Active Geometric Inspection*. Submitted to ICLR 2027.

RESEARCH EXPERIENCE

Learning Physical Responses and Gradients with Differentiable Simulation

Ongoing research | Manuscript in preparation

- Investigate neural surrogate models for predicting physical responses and sensitivities from high-fidelity finite element simulations and adjoint gradients.
- Generate parameterized FEM stress-response datasets and compare value-only learning with direct gradient supervision, including multi-head prediction of objective values, gradient magnitudes, and signs.
- Preliminary experiments show 97.4% material-gradient sign agreement, versus 86.9% for a value-only model differentiated through autograd. Experiments and evaluation are ongoing.

Differentiable Simulation with PolyFEM

Ongoing research infrastructure development

- Developed a schema-driven Python API connecting simulation configurations to the PolyFEM C++ solver; validated generated configurations against the backend.
- Developed structured outputs for mesh, displacement, stress, strain, and von Mises fields, alongside workflows for parameter sweeps and simulation dataset generation.
- Developed adjoint-gradient workflows for differentiable simulation; exploring PyTorch-compatible interfaces for shape and material optimization.

GLANCE | Active Geometric Inspection for 3D Object Identification

Co-author · Manuscript submitted to ICLR 2027

- Contributed to 3D object review, filtering, dataset preparation, and human annotation for a geometry-only object to the GLANCE paper on free-form 3D object identification through active geometric inspection, focusing on benchmark preparation, human annotation, and evaluation.
- Reviewed and filtered 3D objects and contributed to a benchmark of 1,118 shapes, with four independent human annotations per shape.
- Evaluated model predictions against human reference labels, accounting for multiple plausible object identities.

TEACHING EXPERIENCE

Teaching Assistant | CSC 305: Introduction to Computer Graphics

University of Victoria

May 2026 – Present

- Lead CSC 305 lab sessions, explaining computer graphics concepts and guiding students through programming assignments.
- Help students debug graphics implementations and connect algorithms to their code in both in-person and remote labs.
- Grade assignments and exams, provide feedback, and coordinate with the instructor and other teaching assistants on grading consistency.

SELECTED INDUSTRY EXPERIENCE

QA Analyst – Test Automation Engineer Co-op

BC Liquor Distribution Branch, Burnaby, BC

May 2024 – December 2024

- Designed test automation architecture and regression, smoke, and end-to-end testing workflows for the CARM project using C#/.NET, Selenium, SpecFlow, and Eggplant Functional.
- Developed reusable automation libraries and investigated test failures through logs and root-cause analysis to improve maintainability and test reliability.

Software Developer Co-op

Eddyfi Technologies, Nanaimo, BC

September 2023 – April 2024

- Developed C#/.NET desktop applications with WPF, database access, and reporting features; improved large-dataset handling through data virtualization.
- Built a C#/WPF robot testing plugin for repeatable hardware validation and debugged application, data-binding, and database issues.

SELECTED SOFTWARE PROJECTS

Mood Dump | AI-Assisted Journaling and Mood Tracking

- Developed a React Native/Expo app combining private journals, mood tracking, anonymous community sharing, and AI-generated reflections, with Supabase access controls and the OpenAI API.

FlashCall | Automated Phone Assistant

- Built a Node.js system that schedules customer-service calls, navigates phone menus, detects live agents through speech recognition, and transfers calls to users using Twilio and Google Speech-to-Text.

TECHNICAL SKILLS

Scientific computing and ML: Python, C++, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, MATLAB; finite element simulation and adjoint-gradient workflows.

Software development: C#, C, Java, JavaScript, TypeScript, SQL; .NET, WPF, React, React Native, Node.js.

Tools and platforms: Git, GitHub, Docker, AWS, Jupyter Notebook; Oracle SQL, SQL Server, SQLite, Supabase; Selenium and SpecFlow.

CERTIFICATIONS

Machine Learning Specialization — DeepLearning.AI

May 2025

Google Cybersecurity Specialization — Google

July 2025

AWS Cloud Technology Consultant Specialization — AWS

May 2025